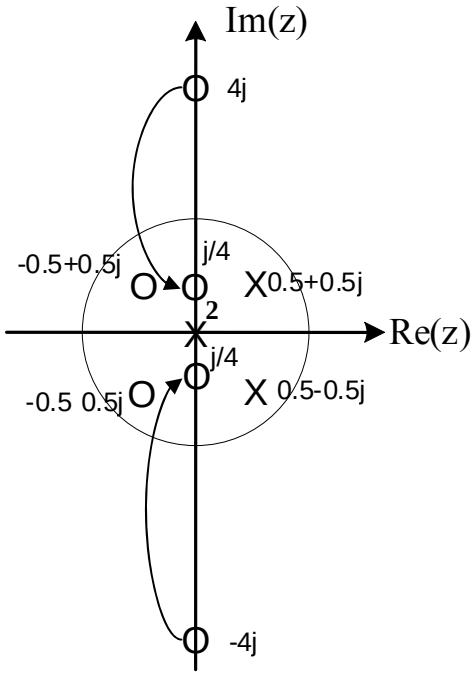
Sample Final Exam Paper 2
Model Answers

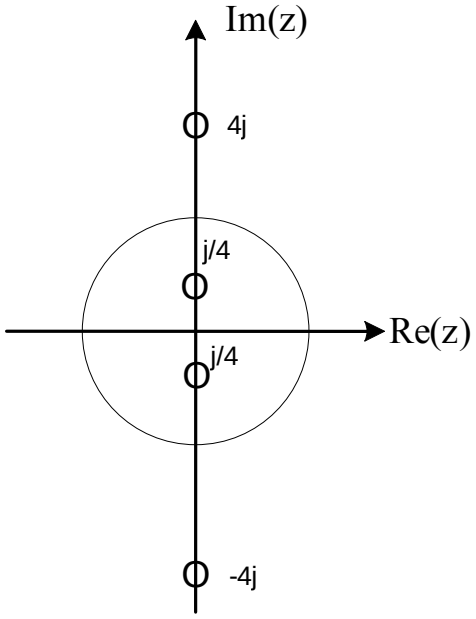
Question 1
(A)
Minimum phase



$$H_l(z) = \frac{(z - j/4)(z + j/4)(z + 1/2 - j/2)(z + 1/2 + j/2)}{(z - 1/2 - j/2)(z - 1/2 + j/2)z^2}$$

$$= \frac{(z - j/16)\left((z + 1/2)^2 + 1/4\right)}{\left((z - 1/2)^2 + 1/4\right)z^2}$$

All pass



$$H_{ap}(z)=\frac{(z-4)(z+4)}{(z-j/4)(z+j/4)}$$
$$\therefore H(z)=H_l(z)H_{ap}(z)$$

(B)

(i)

$$H(z)=a_0+\frac{1}{3}z^{-1}+a_2z^{-2}$$

$$H(\theta)\big|_{\theta=0}=1=a_0+\frac{1}{3}+a_2$$

For linear phase $a_0=a_2$

$$1=a_0+\frac{1}{3}+a_0$$

$$\therefore a_0=\frac{1}{3},\quad a_2=\frac{1}{3}$$

(C)

(i)

$$R(z)=X(z)+z^{-1}Y(z)(-r\sin\theta_0)+r\cos\theta_0R(z)z^{-1}$$
$$R(z)(1-r\cos\theta_0z^{-1})=X(z)-r\sin\theta_0z^{-1}Y(z)\quad ----(1)$$
$$Y(z)=R(z)r\sin\theta_0z^{-1}+Y(z)r\cos\theta_0z^{-1}$$
$$R(z)=\frac{Y(z)[1-r\cos\theta_0z^{-1}]}{r\sin\theta_0z^{-1}}\quad -----(2)$$

From (1) and (2):

$$\frac{X(z)-r\sin\theta_0z^{-1}Y(z)}{1-r\cos\theta_0z^{-1}}=\frac{Y(z)[1-r\cos\theta_0z^{-1}]}{r\sin\theta_0z^{-1}}$$
$$X(z)r\sin\theta_0z^{-1}-r^2\sin^2\theta_0z^{-2}Y(z)=Y(z)[1-2r\cos\theta_0z^{-1}+r^2\cos^2\theta_0z^{-2}]$$
$$\therefore\frac{Y(z)}{X(z)}=\frac{r\sin\theta_0z^{-1}}{1-2r\cos\theta_0z^{-1}+r^2\theta_0z^{-2}}$$

(ii)

$$\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2\theta_0 z^{-2}}$$

(D)

$$H(z) = \left(1 - z^{-1}\right)^3$$

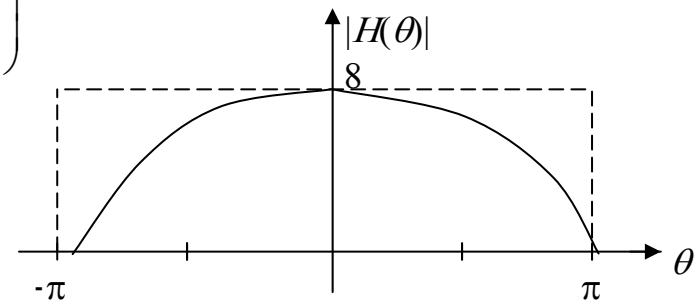
$$H(\theta) = \left(1 - e^{-j\theta}\right)^3$$

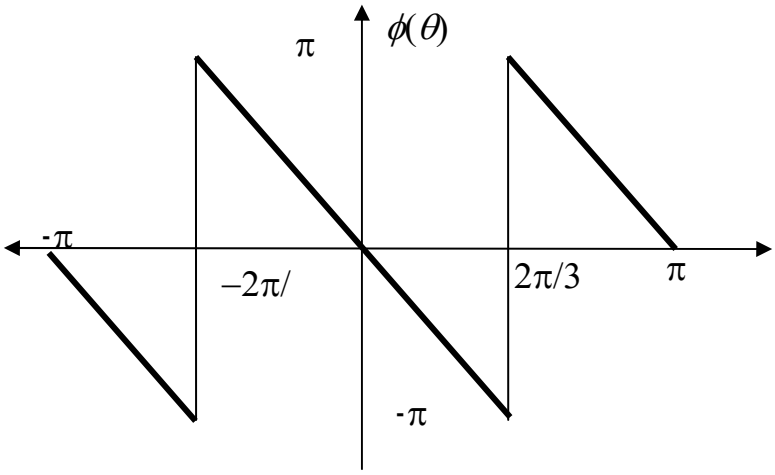
$$= \left(-e^{-j\frac{\theta}{2}}\right)^3 \left(e^{j\frac{\theta}{2}} + e^{-j\frac{\theta}{2}}\right)^3$$

$$= e^{-j\frac{3\theta}{2}} \times 8 \times \cos^3\frac{\theta}{2}$$

$$\therefore |H(\theta)| = \left|8\cos^3\frac{\theta}{2}\right|$$

$$\therefore \phi(\theta) = -\tfrac{3}{2}\theta$$





(E)

$$H(\theta) = \frac{k_0^2}{\left(1 - b_1 e^{-j\theta}\right)^2}$$

$$H(\theta)\big|_{\theta=0} = 1 \quad \Rightarrow \quad 1 = \frac{k_0^2}{(1-b_1)^2} \quad \Rightarrow k_0^2 = (1-b_1)^2$$

$$|H(\theta)|^2 = \frac{k_0^4}{\left(1 + b_1^2 - 2b\cos\theta\right)^2}$$

$$|H(\theta)|^2_{\theta=\frac{\pi}{4}} = \tfrac{1}{4} = \frac{k_0^4}{\left(1 + b_1^2 - 2b\times\frac{1}{2}\right)^2}$$

$$4(1-b_1)^4 = (1 + b_1^2 - b_1)^2$$

Taking square root on both side:

$2(1-b_1)^2 = 1+b_1^2 - b_1 \quad - \quad eqn(1)$

$2(1-b_1)^2 = -(1+b_1^2 - b_1) \quad - \quad eqn(2)$

From eqn(1):

$b_1 = 0.38 \quad or \quad b_1 = 2.62$

but $b_1 = 2.62$ is >1 → unstable system

$\therefore b_1 = 0.38$

$\therefore k_0 = 1 - b_1 = 0.6180$

From eqn(2): solving the quadratic equation results in an imaginary part in the solution so it's invalid.

Question 2

(A)

(i)

From the 1st equation:

$$\cos(n+1)\theta = 2\cos\theta + \cos n\theta - \cos(n-1)\theta$$

From the 2nd equation:

$$\sin n\theta = \frac{\cos(n+1)\theta - \cos(n-1)\theta}{-2\sin\theta}$$

Let $y_1(n) = \cos n\theta$

$y_2(n) = \sin n\theta$

$$y_1(n+1) = 2\cos\theta y_1(n) - y_1(n-1) \quad ----(1)$$

$$y_2(n) = \frac{1}{2\sin\theta}\left[y_1(n-1) - y_1(n+1)\right] \quad ----(2)$$

(ii) Draw the block diagram from the above calculation.

(iii)

$\theta = \frac{3}{4}\pi$

$\therefore a = 2\cos\frac{3}{4}\pi = -= \sqrt{2}$

$$b = \frac{1}{2\sin\frac{3}{4}\pi} = \frac{\sqrt{2}}{2}$$

B

(i)

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

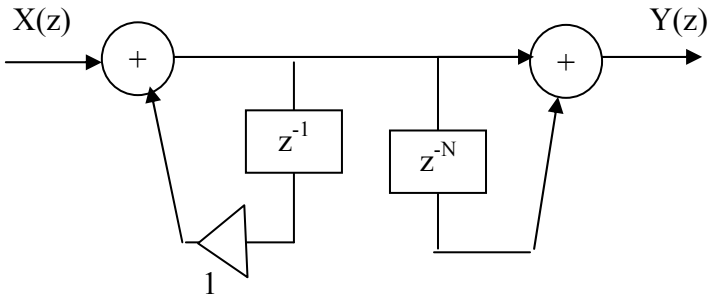
$$H(z) = \frac{1 - z^{-5}}{1 - z^{-1}}$$

$$H(\theta) = \frac{1 - e^{-j5\theta}}{1 - e^{-j\theta}} = \frac{e^{-j\frac{5}{2}\theta}}{e^{-j\frac{1}{2}\theta}} \frac{\left(e^{j\frac{5}{2}\theta} - e^{-j\frac{5}{2}\theta}\right)}{\left(e^{-j\frac{1}{2}\theta} - e^{-j\frac{1}{2}\theta}\right)}$$

$$H(\theta) = e^{-j2\theta} \frac{\sin\frac{5}{2}\theta}{\sin\frac{1}{2}\theta}$$

(ii)

$$H(z) = \frac{1 - z^{-5}}{1 - z^{-1}}$$



(iii)

$$\phi(\theta) = -2\theta$$

$$\therefore \tau_g = -\frac{d\phi(\theta)}{d\theta} = 2 \quad \text{(constant group delay)}$$

Question 3:

(A)

$$x[n]=\begin{cases}1 & -N\le n\le N\\ 0 & otherwise\end{cases}$$

$$Let \quad X(\theta)\xleftrightarrow{DTFT}x[n]$$

$$X(\theta)=\sum_{n=-\infty}^{\infty}x[n]e^{-j\theta n}=\sum_{n=-N}^{N}e^{-j\theta n}$$

$$Let \quad m=n+N$$

$$X(\theta)=\sum_{m=0}^{2N}e^{-j\theta(m-N)}=e^{j\theta N}\sum_{m=0}^{2N}e^{-j\theta m}$$

$$=e^{j\theta N}\left[\frac{1-e^{-j\theta(2N+1)}}{1-e^{-j\theta}}\right]$$

$$=\frac{e^{j\theta N}-e^{-j\theta N}e^{-j\theta}}{1-e^{-j\theta}}$$

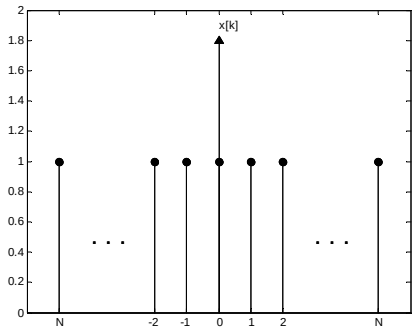
$$=\frac{e^{j\theta N}e^{j\frac{\theta}{2}}-e^{-j\theta N}e^{-j\frac{\theta}{2}}}{e^{j\frac{\theta}{2}}-e^{-j\frac{\theta}{2}}}$$

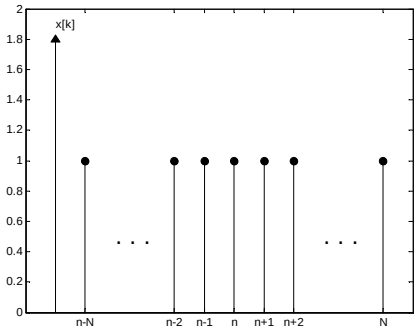
$$=\frac{e^{j\theta\left(N+\frac{1}{2}\right)}-e^{-j\theta\left(N+\frac{1}{2}\right)}}{e^{j\frac{\theta}{2}}-e^{-j\frac{\theta}{2}}}$$

$$=\frac{\sin\left[\left(N+\frac{1}{2}\right)\theta\right]}{\sin\frac{\theta}{2}}=\frac{\sin\left[(2N+1)\frac{\theta}{2}\right]}{\sin\frac{\theta}{2}}$$

(B)

$$x[n]*x[n]=\sum_{k=-\infty}^{\infty}x(k)x(n-k)$$





$For\ n < -2N \quad x(k)x(n-k) = 0 \quad \forall k$

$For\ -2N \le n \le 0 \quad x(k)x(n-k) = \begin{cases} 1 & -N < k \le n+N \\ 0 & otherwise \end{cases}$

$For\ 0 \le n \le 2N \quad x(k)x(n-k) = \begin{cases} 1 & n-N < k \le N \\ 0 & otherwise \end{cases}$

$For\ n > 2N \quad x(k)x(n-k) = 0 \quad \forall k$

$\therefore x[n] * x[n]$

$$= \begin{cases} \sum_{k=-N}^{n+N} 1 & -2N \le n \le 0 \\ \sum_{k=n-N}^{N} 1 & 0 \le n \le 2N \end{cases}$$

$$= \begin{cases} 2N+n & -2N \le n \le 0 \\ 2N-n & 0 \le n \le 2N \end{cases}$$

$= y[n]$

$Let \quad Y(\theta) \xleftrightarrow{DTFT} y[n]$

$Y(\theta) \xleftrightarrow{DTFT} x[n] * x[n]$

$\therefore Y(\theta) = X(\theta)X(\theta) \quad (\text{Convolution property})$

Question 4:

(A)

$h(n) \xrightarrow{DTFT} H(\theta)$

$v(n) = x(-n) * h(n)$

$u(n) = v(-n) = x(n) * h(n)$

$y(n) = \{h(n) + h(-n)\} * x(n)$

$Y(z) = \{H(z) + H(z^{-1})\} \bullet X(z)$

$G(z) = H(z) + H(z^{-1})$

$G(\theta) = H(\theta) + H^*(\theta) = (a + jb) + (a - jb) = 2a$

Hence the equivalent transfer function $G(\theta)$ is real and has zero phase.

(B)

$$H(s)=\frac{0.1s}{s^2+0.1s+1}$$

$$\theta_C=2\pi\frac{fc}{fs}=\frac{\pi}{2}$$

$$\therefore \omega'_C=\frac{2}{T}\tan\left(\frac{\theta_C}{2}\right)=\frac{2}{T}\ \text{rad/sec}$$

Pre-warped analogue filter:

$$H(s)=\frac{0.1\times\frac{s}{\frac{2}{T}}}{\left(\frac{s}{\frac{2}{T}}\right)^2+0.1\times\frac{s}{\frac{2}{T}}+1}$$

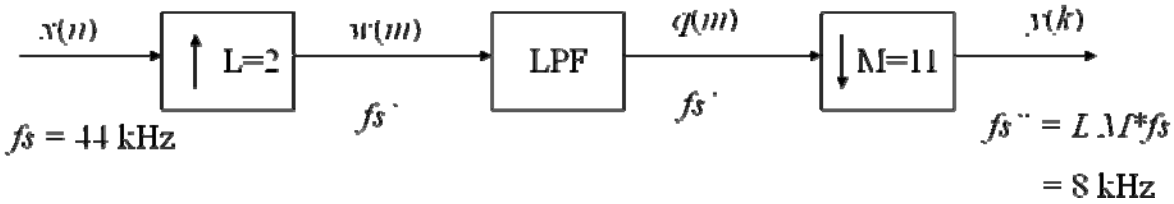
$$H(s)\big|_{s=\frac{2}{T}\frac{z-1}{z+1}}=H(z)=\frac{0.1\times\frac{2}{T}\times\frac{2}{T}\frac{z-1}{z+1}}{\left(\frac{2}{T}\right)^2\left(\frac{z-1}{z+1}\right)^2+0.1\times\frac{2}{T}\times\frac{2}{T}\frac{z-1}{z+1}+\left(\frac{2}{T}\right)^2}$$

$$H(z)=\frac{1}{21}\frac{z^2-1}{z^2+\frac{19}{21}}=\frac{1}{21}\frac{1-z^{-2}}{1+\frac{19}{21}z^{-2}}$$

Question 5:

(A)

(i)



$$\frac{L}{M}=\frac{8000}{44000}=\frac{2}{11}$$

(ii)

Using IIR filter, we cannot take advantage of the fact that we have to compute only every *N* th output, since previous outputs are needed to compute the *N* th output.

On the other hand, using FIR filter we can do our computation at the rate of fs'/N → a significantly lower computational rate.

(B)

$$\theta_1 = 2\pi \times 2 \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2}$$

$$\theta_2 = \omega_C T + \omega_1 T = \frac{\pi}{2} + 2\pi \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$X(\theta) = 1 + \cos\theta$$

$$B_C(\theta) = \frac{1}{2}\left[X(\theta-\theta_1) + X(\theta+\theta_1)\right] = \frac{1}{2}\left[1 + \cos\left(\theta - \frac{\pi}{2}\right) + 1 + \cos\left(\theta + \frac{\pi}{2}\right)\right] = \frac{1}{2}\times 2 = 1$$

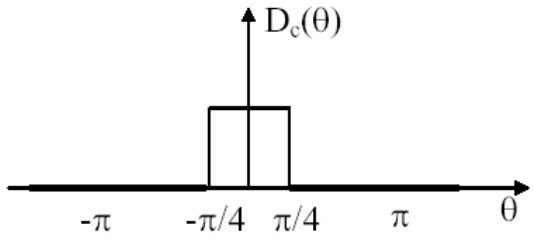
$$D_C(\theta) = B_C(\theta)\cdot G(\theta) = 1, \quad 0 < |\theta| < \tfrac{\pi}{4}$$

$$R_C(\theta) = \frac{1}{2}\left[D_C(\theta-\theta_2) + D_C(\theta+\theta_2)\right] = \frac{1}{2}\left[D_C\left(\theta - \tfrac{3}{4}\pi\right) + D_C\left(\theta + \tfrac{3}{4}\pi\right)\right]$$

$$R_C(\theta) = \begin{cases} \frac{1}{2} & -\frac{\pi}{4} \le \theta - \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\frac{\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow -\pi \le \theta \le -\frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Then from the above calculation, sketch $X(\theta), B_C(\theta), D_C(\theta)$, and $R_C(\theta)$.

$$\mathrm{B_c}(\theta) = 1, \mathrm{D_c}(\theta) = 1$$

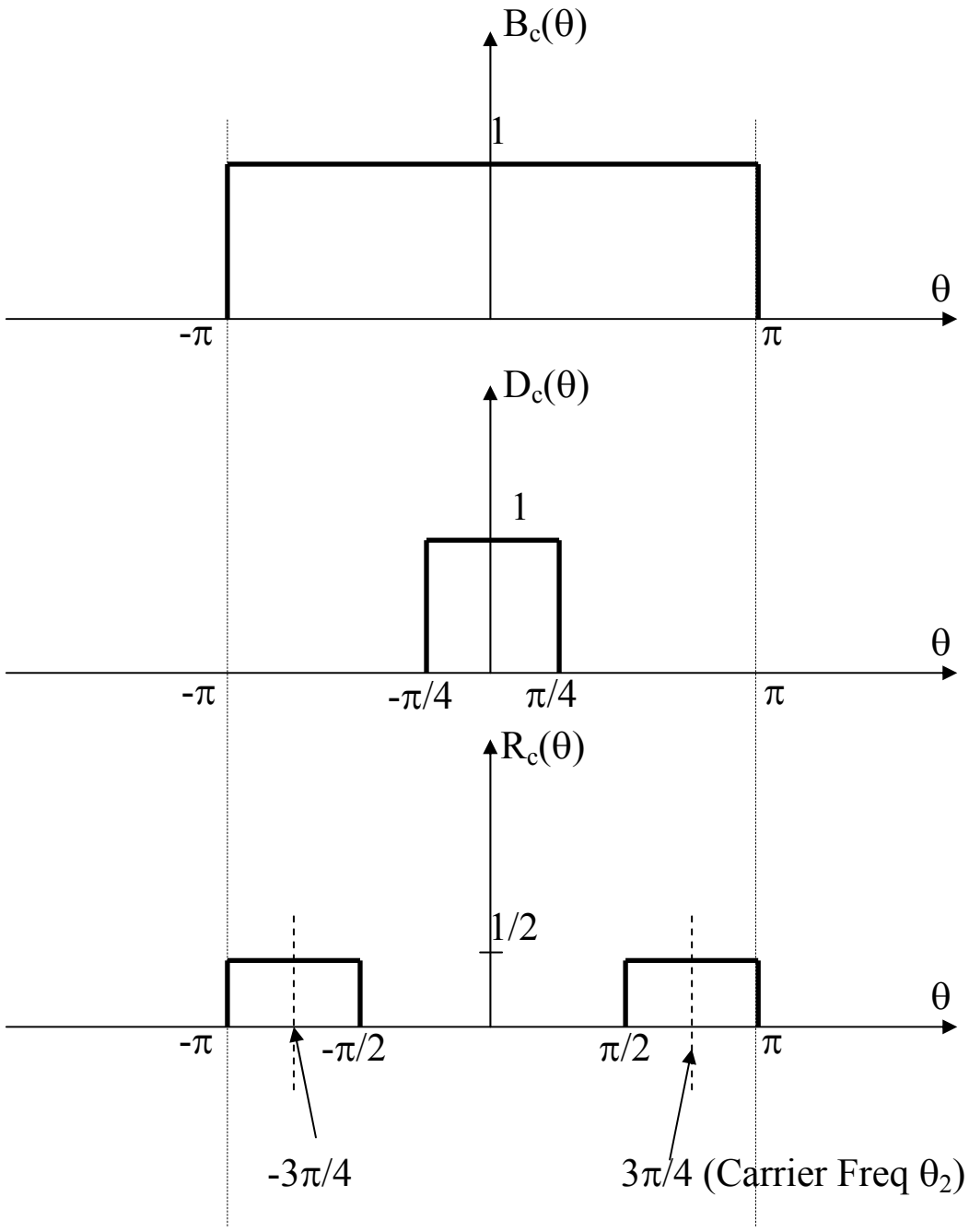


$$\mathrm{R_c}(\theta) = \frac{1}{2}\left\{\mathrm{D_c}(\theta - \theta_1) + \mathrm{D_c}(\theta + \theta_1)\right\}$$

$$= \frac{1}{2}\left\{ D_C(\theta - \frac{3\pi}{4}) + D_c(\theta + \frac{3\pi}{4}) \right\}$$

$$= \begin{cases} \frac{1}{2} & \frac{-\pi}{4}(\theta - \frac{3\pi}{4}) \le \frac{\pi}{4} \\ \\ \frac{1}{2} & \frac{-\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \\ \\ 0 & \text{elsewhere} \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \frac{\pi}{2} \le \theta \le \pi \\ \\ \frac{1}{2} & -\pi \le \theta \le \frac{-\pi}{2} \\ \\ 0 & \text{elsewhere} \end{cases}$$


$B_c(\theta)$
1
θ
-π
π
$D_c(\theta)$
1
θ
-π
-π/4
π/4
π
$R_c(\theta)$
1/2
θ
-π
-π/2
π/2
π
-3π/4
3π/4 (Carrier Freq θ_2)